

A lightweight enhanced YOLO framework for accurate small-target cable defect detection

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A lightweight enhanced YOLO framework for accurate small-target cable defect detection

Guangyi Yang¹³, Fan Zhou², Biaofeng Di², Xiaomin Wu²

Abstract –The reliable detection of cable defects is critical for the safety and stability of power transmission systems. Traditional methods often struggle to identify small or blurred defects efficiently. This study presents YOLO-ESBD, an enhanced YOLO11-based framework for cable defect detection.

The model integrates an optimized feature fusion strategy, an effective attention mechanism, and a lightweight detection head, improving the representation of small targets while maintaining real-time performance. Experiments on a specialized cable defect dataset show that YOLO-ESBD achieves a precision of 92.3%, recall of 92.1%, and mAP50 of 94.2%, with a model size of 5.9 MB and computational cost of 9.2 GFLOPs. These results indicate superior accuracy, robustness, and efficiency compared to mainstream algorithms. YOLO-ESBD demonstrates strong potential for practical deployment in an automated power inspection, supporting intelligent monitoring and preventive maintenance of cable networks.

Keywords: Cable Defect Detection, YOLO-ESBD, Deep Learning, Power Transmission Safety, Neural Networks

1. Introduction

With the rapid development of high-quality economy and the accelerated construction of new power systems, the demand for electricity transmission has been continuously increasing. As a core carrier in power transmission, distribution, and conversion, the performance and reliability of cables are directly related to the safety of power infrastructure and the nor-

mal operation of the national economy [1]. However, during cable manufacturing, surface defects such as wrinkles, damages, incomplete sheath removal, and foreign matter attachment frequently occur [2]. If undetected, these defects may cause equipment failures, production downtime, and economic losses [3], while in operation, they can compromise insulation and mechanical strength, potentially leading to partial discharge, short circuits, or even fire hazards, threatening grid safety and personnel safety [4], and increasing operation and maintenance costs [5].

Cable surface defects are usually small, complex, and randomly distributed. Combined with high-speed production lines, traditional manual visual and tactile inspection is inefficient, subjective, and inadequate for comprehensive, precise, and real-time quality control [6]. Cu-

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urrent cable defect detection methods mainly include non-contact and contact techniques. Non-contact approaches such as infrared thermography, ultrasonic, optical imaging, and electromagnetic detection enable fast, flexible, and non-destructive detection [7]. Infrared thermography is effective for bulges and pre-breakdown defects [8][9], while ultrasonic and electromagnetic methods are suitable for cracks and peeling, yet they are sensitive to environmental disturbances and exhibit limited accuracy under complex backgrounds. Contact methods, including partial discharge detection, X-ray imaging, coating peel tests, and pulse-thermal inspection, offer high precision and sensitivity to subtle defects [10][11], but are complex, low in efficiency, potentially damaging to cables, and unsuitable for online industrial deployment.

Recently, deep learning and machine vision have been increasingly applied to cable defect detection [12][13]. For instance, ViTR Net [14] combines CNN and Transformer for lightweight unsupervised defect detection using only normal cable images, but generalization to unseen defects is limited. The improved YOLOv8n [15] simultaneously detects cable bending and defects, enhancing small target recognition, but relies on extensive annotated data and increases model complexity. Improved YOLOv5s [16] and FC-YOLO [17] enhance detection through feature fusion and attention mechanisms, though at the cost of higher computational overhead and deployment difficulty. YOLO-CSM [18] integrates dual attention mechanisms and additional detection layers to improve micro-defect detection, but challenges remain for real-time deployment and adaptability in complex environments.

To address limitations in small target detection, complex background handling, and edge-device deployment efficiency, this paper proposes YOLO-ESBD (Efficient Spatial-BiFusion-

Dynamic YOLO). The main innovations include:

- (1) Introducing EMA (Efficient Multi-scale Attention) in the backbone network to enhance multi-scale feature representation;
- (2) Incorporating SPDConv (Spatially Preserving Dilated Convolution) to strengthen shallow feature expression and improve small target perception;
- (3) Replacing the original Concat with BiFPN (Bidirectional Feature Pyramid Network) for efficient multi-scale feature fusion;
- (4) Employing DyHead dynamic detection module in the detection head, enabling adaptive detection strategies based on image region complexity.

These improvements allow YOLO-ESBD to achieve higher precision in small and multi-defect detection while maintaining lightweight design and deployment efficiency, providing an effective and accurate solution for industrial intelligent cable surface defect detection.

2. YOLO-ESBD Model

2.1 YOLO-11 Algorithm

Since its initial release, the YOLO series of object detection algorithms has undergone multiple iterations and optimizations, resulting in various versions. While maintaining an end-to-end structure and high efficiency, YOLO-11 achieves significant improvements in detection accuracy, model lightweighting, and generalization capability, surpassing earlier versions such as YOLOv8, YOLOv9, and YOLOv10. It demonstrates stronger practical application potential, particularly excelling in detecting multiple defects in complex scenarios [19]. Compared with its predecessor, YOLO11 introduces key innovations in network architecture design.

As shown in Figure 1, the original architecture

of YOLO-11 incorporates several core improvements. In the backbone network, the C2f module from YOLOv8 is replaced with the novel C3k2 module, effectively combining the advantages of both C2f and C3 modules. The C3k2 module optimizes the feature extraction pathway and gradient propagation mechanism, enhancing the model's ability to capture feature information and improving the training stability of deep networks, thereby further boosting detection accuracy and training efficiency. Additionally, to improve recognition of occluded and multi-scale objects in complex environments, YOLO-11 introduces the C2PSA module after the SPPF (Spatial Pyramid Pooling - Fast) layer in the backbone. This module employs a dynamic attention mechanism to strengthen local and global feature modeling, enhancing the model's adaptability and robustness across diverse scenarios.

In terms of overall design, YOLO-11 retains the PGI (Path-Guided Information) architecture first introduced in YOLOv9, which constructs optimized gradient propagation paths to mitigate information attenuation in deep networks. YOLO-11 also preserves the dual-label assignment mechanism proposed in YOLOv10, directly generating final prediction boxes and bypassing the conventional NMS (Non-Maximum Suppression) process. This simplifies both training and deployment while improving inference speed and detection efficiency.

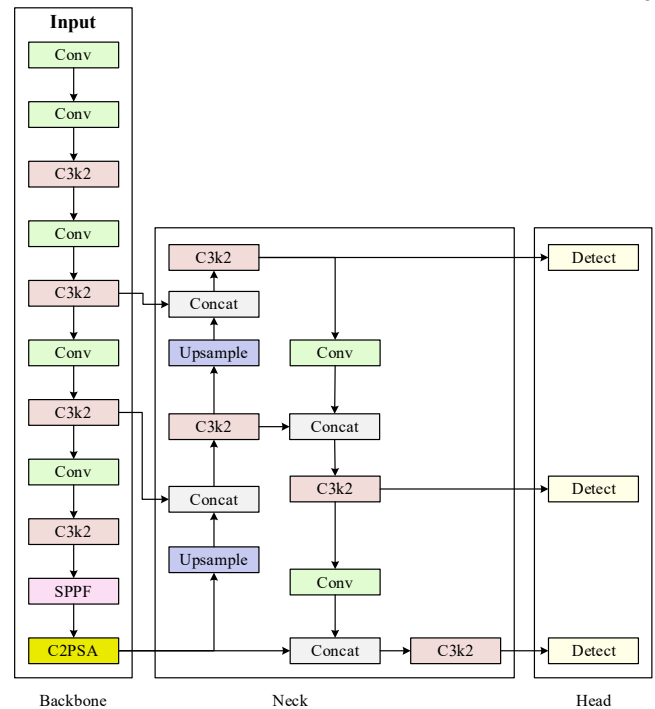


Fig. 1. Original YOLO11 architecture.

The YOLO-11 network is composed of four main components: the input layer, backbone, neck, and detection head. The input layer performs adaptive scaling, normalization, and data distribution alignment to ensure consistency between input features and the training distribution, thus enhancing generalization. The backbone extracts multi-level semantic features from the input image and propagates critical information through deep convolutional structures. Within this backbone, the C3k2 and C2PSA modules are key components, jointly optimizing the richness and expressiveness of feature representations. The neck acts as an intermediary connecting the backbone and detection head, further integrating feature maps at multiple scales. In this module, operations such as C3k2, upsampling, convolution, and feature concatenation (Concat) are employed to fully fuse multi-scale information, improving detection performance for small targets and objects in complex backgrounds. Finally, the detection head is divided into multiple branches to perform classification, localization, and confidence

estimation. Carefully designed loss functions measure and optimize the discrepancy between model predictions and ground truth, ensuring convergence and accuracy during training.

Through comprehensive innovations in network architecture, module design, and

optimization mechanisms, YOLO-11 not only enhances the accuracy and real-time performance of object detection but also provides a solid foundation for its application in industrially demanding tasks such as cable defect recognition and small-object detection.

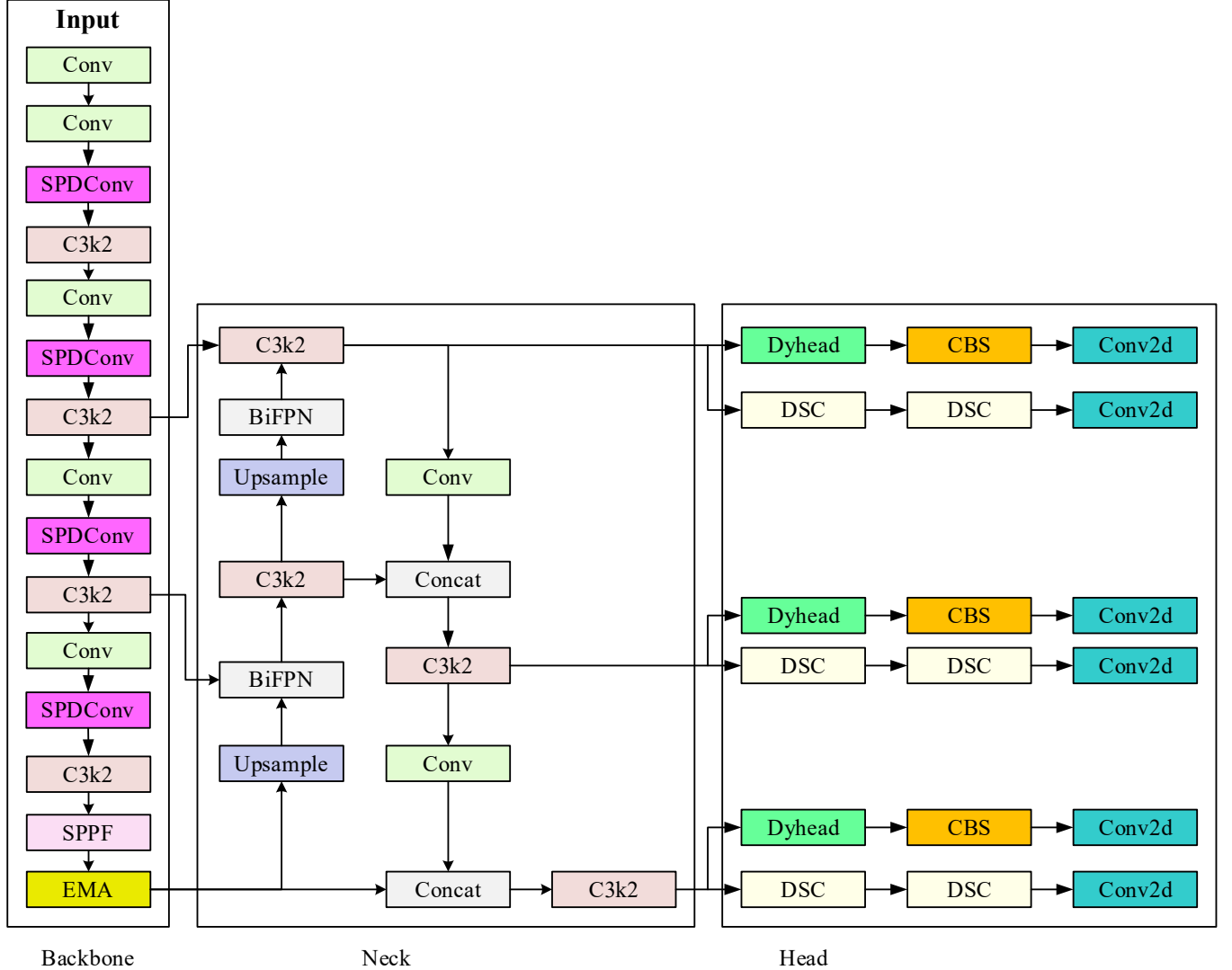


Fig. 2. Proposed YOLO-ESBD improved architecture.

2.2 YOLO-ESBD Network Architecture

Based on the YOLO11 model, this study proposes a multi-module integrated improved model, named YOLO-ESBD (Efficient Spatial–BiFusion–Dynamic YOLO), aimed at enhancing detection performance for power cable surface defects in complex environments. In the

backbone network, the original C2PSA attention mechanism is replaced with the EMA (Efficient Multi-scale Attention) mechanism to more efficiently capture multi-scale contextual information and strengthen the model’s focus on target regions. Meanwhile, the SPDConv structure is introduced to preserve spatial consistency of shallow features, further

improving the perception of small objects. During the feature fusion stage, YOLO-ESBD employs the BiFPN structure instead of the traditional Concat operation, using its learnable bidirectional fusion paths to efficiently integrate features at different scales and enhance semantic information propagation. In the detection head, the DyHead dynamic detection module replaces the static CBS structure, combining dynamic attention across spatial, scale, and task levels to effectively improve localization accuracy and classification robustness for multi-type defect targets. These four improvements collaboratively enhance the balance between detection accuracy, generalization capability, and real-time performance of the YOLO-ESBD model, while maintaining lightweight characteristics suitable for deployment on edge computing devices. The following subsections provide detailed descriptions of the design principles and technical details of each key module. The improved network architecture is shown in Figure 2.

2.3 EMA Attention Mechanism

The attention mechanism, as a key architectural component in deep learning models, enhances the selective emphasis of features, enabling the model to focus more on critical image regions while suppressing redundant or irrelevant background information. This significantly improves detection performance and strengthens model generalization. The Efficient Multi-scale Attention (EMA) mechanism is an optimized module designed to address the limitations of traditional multi-scale attention methods. Its core idea is to achieve efficient learning and fusion of cross-channel information while maintaining controllable model complexity [20]. By employing channel reorganization and feature grouping strategies, the input feature map is divided into multiple sub-feature groups (with the number of groups g much smaller than the channel number C , i.e., $g \ll C$)

, thereby extracting rich contextual information at different semantic levels and enhancing the perception of multi-scale targets.

The overall architecture of the EMA module is shown in Figure 3. It integrates multiple functional components, including one-dimensional horizontal and vertical global pooling (denoted as X-average pooling and Y-average pooling, respectively), convolutional operations (Conv), matrix multiplication (Matmul), normalization layers (GroupNorm), a weight redistribution module (Reweight), grouped convolutions (Groups), and activation functions (Sigmoid and Softmax). The asterisk “*” indicates fusion operations, representing feature interactions among the relevant inputs. Within this mechanism, the feature map is first divided into g sub-feature groups, which are processed independently in parallel subnetworks to extract multi-level attention features.

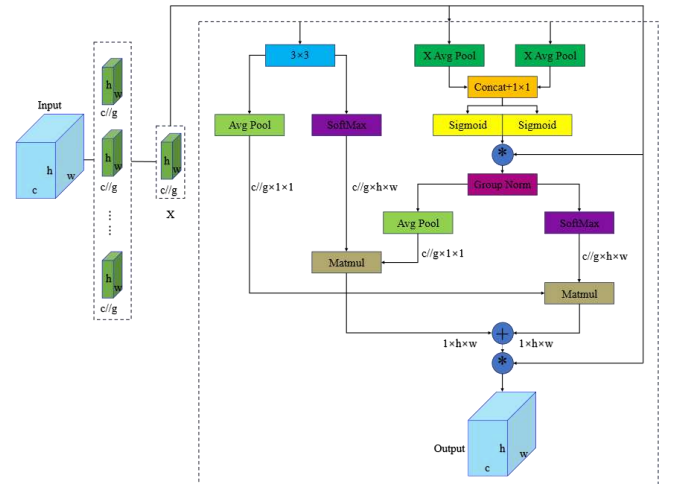


Fig. 3. Structure of the EMA module.

Attention weights are modeled through three parallel branches. Two of them employ 1×1 convolutional branches to strengthen inter-channel information interactions, while the third branch introduces a 3×3 convolution to capture stronger local contextual features, expanding the receptive field across channels. Particularly in the 1×1 convolutional branches, the EMA module incorporates two-dimensional global average pooling to compactly model cross-channel interactions within each group. The resulting

channel attention maps are aggregated through multiplication, achieving effective cross-channel fusion. In contrast, the 3×3 convolutional branch focuses on modeling short-range dependencies between local channels, further enhancing spatial feature representation.

In addition, the EMA mechanism incorporates cross-spatial learning strategies, enabling the integration of multidimensional contextual information over a broader range and constructing a more comprehensive and fine-grained feature representation. This joint modeling of spatial and channel dimensions significantly strengthens the model's feature extraction capability, making it more robust and adaptable to multi-scale object detection tasks.

As an efficient and structurally optimized multi-scale attention mechanism, EMA leverages channel reorganization, feature grouping, parallel convolutional paths, and cross-spatial information fusion to markedly improve the ability of deep networks to model key target regions. While keeping computational complexity low, it effectively enhances detection performance in complex scenarios.

2.4 SPD-Conv Block

When addressing the tasks of detecting low-resolution images of cable surface defects and small objects in this study, the original YOLO11 model exhibits certain limitations. Its detection performance tends to deteriorate significantly when the target size is small or the image resolution is constrained, which is particularly evident in practical applications such as rail defect detection. These tasks often involve numerous tiny and easily confusable defect regions, whereas the original YOLO11 model lacks sufficient capability in feature extraction and fine-grained information preservation, thereby restricting its detection accuracy. To

overcome these limitations, this paper introduces a structural optimization strategy by integrating an SPD-Conv (Sparse Pointwise Downsampling Convolution) module after each convolutional layer of the YOLO11 model, thereby enhancing its adaptability and detection performance for small-object and low-resolution image scenarios.

$$\begin{aligned} G_{0,0} &= X[0:S:scale, 0:S:scale] \\ G_{1,0} &= X[1:S:scale, 0:S:scale] \\ G_{0,1} &= X[0:S:scale, 1:S:scale] \\ G_{1,1} &= X[1:S:scale, 1:S:scale] \\ X' &= G_{0,0} \parallel G_{1,0} \parallel G_{0,1} \parallel G_{1,1} \end{aligned} \quad (1)$$

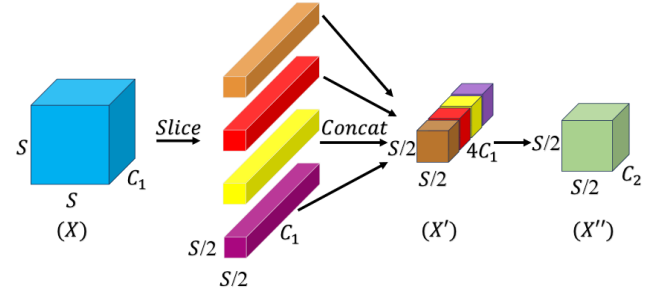


Fig. 4. SPD-Conv module structure.

The SPD-Conv module is composed of an SPD sparse pointwise downsampling layer and a non-strided convolutional layer. Its central idea is to achieve spatial downsampling in a more efficient way while preserving the completeness of feature information [21]. Unlike traditional strided convolution or pooling operations that directly reduce spatial dimensions, the SPD layer adopts a sparse sampling strategy, which avoids discarding important feature information and thereby enhances the ability to retain fine details.

As shown in Fig. 4 and Eq. (1), for an intermediate feature map with size $S \times S \times C_1$, the SPD operation with a downsampling scale of 2 divides it into four non-overlapping sub-feature maps G_1 , G_2 , G_3 , and G_4 , each of size $(S/2) \times (S/2) \times C_1$. These sub-feature maps are then concatenated along the channel dimension to form a new feature map X' with size $(S/2) \times (S/2) \times 4C_1$. In this way, spatial compression is achieved while structural

information is preserved through channel expansion. This process significantly enhances the model's representational ability under low-resolution images and is particularly suitable for the extraction and recognition of small objects.

After the SPD operation, a non-strided convolutional layer, that is, a standard convolution with stride = 1, is applied to further extract local features and reorganize the representation. This convolutional layer employs C_2 filters, where $C_2 < 4C_1$, with the aim of reducing the channel dimension while retaining key discriminative information. As a result, an output feature map X'' of size $(S/2) \times (S/2) \times C_2$ is generated. This configuration effectively mitigates information loss without introducing additional parameter overhead, thereby providing a solid foundation for subsequent feature fusion and object localization.

By incorporating the SPD-Conv module into each convolutional stage of YOLO11 and eliminating conventional strided convolution and pooling operations, the overall architecture achieves stronger feature retention and improved sensitivity to small objects. Consequently, when handling low-quality input images commonly encountered in practical cable production scenarios, the enhanced model demonstrates greater robustness and higher detection accuracy.

2.5 BiFPN Block

The traditional Feature Pyramid Network (FPN) employs a top-down structural design to construct a series of feature maps with uniform channel dimensions but varying spatial resolutions. This design endows the model with strong multi-scale feature representation capabilities, enabling effective detection of objects of different sizes. However, in practical applications, FPN suffers from several limitations, including unidirectional information flow, shallow feature fusion, and restricted efficiency in feature transmission. These issues particularly

constrain its performance when dealing with small objects or low-contrast targets. To address these challenges, the Bidirectional Feature Pyramid Network (BiFPN) introduces a more efficient bidirectional feature fusion mechanism. As illustrated in Fig. 5, BiFPN retains the multi-scale processing capability of the original FPN while incorporating bidirectional information flow paths. This allows features to propagate and fuse simultaneously in both top-down and bottom-up directions, fundamentally overcoming the limitations of restricted information flow inherent in the FPN structure [22].

Through iterative bidirectional fusion, BiFPN continuously strengthens the interaction depth and semantic correlation among cross-scale features, effectively alleviating redundancy and repetition in information transmission while enhancing the compactness and discriminability of feature representations. In addition, to further improve fusion quality, BiFPN introduces a weighted feature fusion strategy with learnable parameters, which enables the network to adaptively adjust the contribution of each input feature according to its importance in the fusion process. The weighted fusion is formulated as:

$$O = \sum_i \frac{w_i I_i}{\varepsilon + \sum_j w_j} \quad (2)$$

I_i represents the input feature and O represents the output feature. The learnable weights w_1 and w_2 are used to adjust the contribution of different features during fusion, and $\varepsilon = 0.0001$ is included to prevent numerical instability. This weighted fusion mechanism not only enhances the flexibility and expressive power of feature integration but also allows the model to automatically emphasize critical features while suppressing redundant background information. Such capability is particularly beneficial in scenarios with complex lighting, occlusion, or noise interference, significantly

improving detection robustness and accuracy.

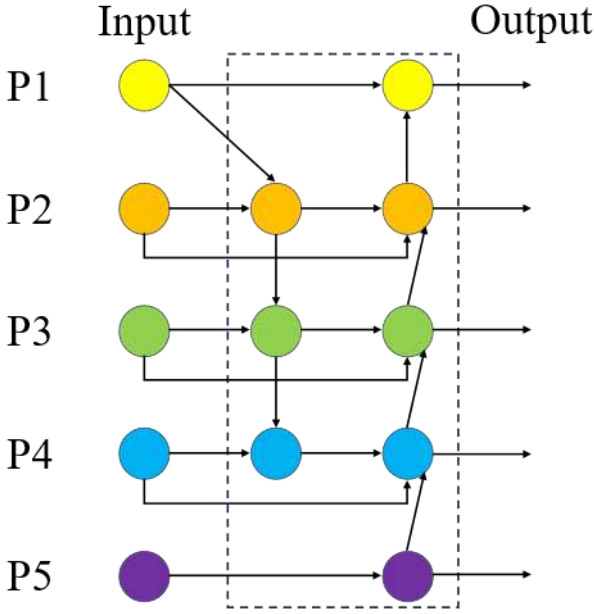


Fig. 5. BiFPN module structure.

In the task of insulator defect detection, BiFPN maintains structural consistency and spatial resolution across multi-scale features, effectively guiding the retention and propagation of small-object information through the feature flow. This approach prevents the dilution and loss of small-object features that commonly occur in traditional architectures due to repeated downsampling, thereby achieving superior detection performance and generalization capability under diverse scales, complex backgrounds, and interference conditions.

2.6 DyHead Module

To improve the accuracy and efficiency of cable surface defect detection, this study introduces the DyHead (Dynamic Head) framework in the detection head, replacing the traditional detection head structure in the original YOLOv11 network. DyHead integrates the object detection task with a multi-dimensional attention

mechanism, forming a detection head with dynamic perception capability. This design not only enhances feature representation but also significantly improves the model's adaptability and detection accuracy across different scenarios [23]. Its core advantage lies in the structured integration of multiple attention mechanisms, enabling deep modeling across semantic levels, spatial positions, and task output dimensions, all without incurring additional computational overhead, thus maintaining the network's efficiency and practical applicability.

As shown in Fig. 6, the DyHead framework integrates three complementary attention mechanisms, each operating on a different feature dimension: L, S, and C. Layer Attention (L) models semantic features across layers, enabling the perception of object information at different scales and thereby enhancing the detection of small objects and multi-scale defects. Spatial Attention (S) focuses on spatial distribution features, encoding the responsive regions of the feature map in space, which allows the model to more accurately localize defect positions. Channel Attention (C) models semantic correlations among channels, emphasizing critical channel information while suppressing irrelevant or redundant features, thus improving discriminative capability. These three attention modules collectively form the basic structural unit of DyHead—the DyHead Block. Multiple such blocks can be stacked in a specific manner to construct a complete detection head module.

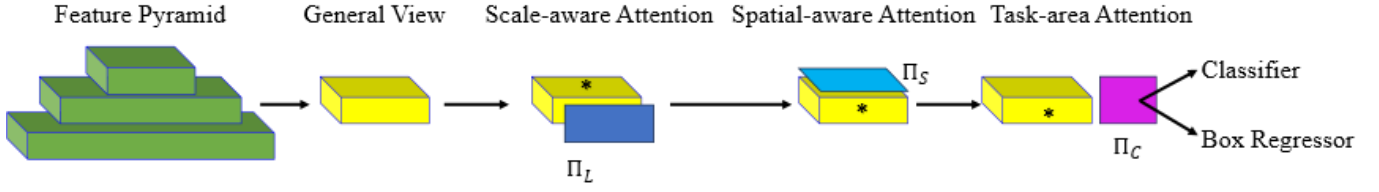


Fig. 6. DyHead design.

Unlike traditional cascaded attention modules, the DyHead implementation used in this study employs only a single DyHead block with a well-designed structure rather than a complex multi-layer stacking. This design significantly reduces parameter redundancy and computational complexity while ensuring efficient execution of the attention mechanisms. By introducing DyHead as the detection head, the model demonstrates enhanced robustness and accuracy when handling subtle cable surface defects, non-uniform textures, and complex backgrounds, validating its practical potential and advantages in industrial visual inspection tasks.

3. Experimental Validation

3.1 Dataset Collection

Our team conducted a study on cable surface defect data collection in the actual production environment of a cable manufacturing plant. Through long-term observation and recording of the production line, we identified four typical defect types that occur most frequently and have the greatest impact on production: wrinkles, breakages, unstripped insulation, and machine scratches. During data acquisition, each defect type was captured on-site using high-resolution cameras, ensuring the authenticity and representativeness of the dataset. Ultimately, this study constructed a cable surface defect dataset consisting of approximately 2,000 original images in JPG format. Each image has a resolution of 1440×2560 pixels, clearly presenting the fine details of the defects and providing a solid data foundation for the subsequent training of intelligent detection models.



Fig. 7. Four types of cable surface defects: wrinkle, breakage, unstripped insulation, and machine scratch.

As shown in Fig. 7, the four defect types in the dataset each possess distinct visual characteristics, while also presenting several detection challenges. (1) The defects are relatively small, occupying only a limited proportion of the overall cable surface, which increases the difficulty for detection models to capture local features. (2) The differences between defects and surrounding normal regions in terms of color, texture, and other visual cues are subtle, which may lead to false positives or missed detections. Moreover, the actual production environment presents a complex background, including production line equipment, conveyor belts, reflective light sources, and other factors, further complicating the extraction of effective features. (3) Although the production equipment is equipped with lighting to ensure image quality, environmental light variations such as glare and shadows remain unavoidable, causing fluctuations in image quality and defect visibility.

The combination of these factors makes cable surface defect detection technically challenging and highlights the limitations of traditional

manual inspection methods in consistently and efficiently meeting production requirements. Additionally, general-purpose computer vision models struggle to achieve satisfactory performance in this specific scenario.

Therefore, constructing a high-quality dataset that closely reflects real production environments not only provides essential data support for training intelligent detection models but also lays the foundation for subsequent research on automated cable surface defect detection. Furthermore, detection models developed for such scenarios need to be highly sensitive to small defects, robust against complex backgrounds, and capable of accurately recognizing defects under varying illumination conditions, imposing higher technical requirements for the application of deep learning and other data-driven methods.

3.2 Data Augmentation and Dataset Splitting

Given the limited number of original cable surface defect images—only 2,000 in total—this study applied data augmentation to improve the generalization capability of the detection model and its adaptability to complex production environments. Spatial augmentations, including rotation, flipping, translation, scaling, cropping, as well as affine and perspective transformations, were used to simulate variations in orientation, position, scale, and viewpoint, thereby enhancing the model's ability to handle diverse imaging conditions and scenarios.

At the color and illumination level, adjustments of brightness, contrast, saturation, and hue, along with the introduction of lighting noise and color jitter, were applied to improve model robustness under varying lighting conditions and imaging settings. To address image perturbations, techniques such as Gaussian noise, salt-and-pepper noise, blurring, and local

occlusion were employed, increasing model stability when noise or partial information loss occurs. Additionally, sample reorganization methods including Mixup and CutMix were used to further enrich the training data distribution.

Advanced augmentation techniques, such as random erasing, Mosaic, style transfer, and generative adversarial network (GAN)-based synthesis, were also applied to increase data diversity and complexity. Automated strategies like AutoAugment and RandAugment were incorporated to reduce reliance on manual selection of augmentation methods. Finally, test-time augmentation (TTA) was employed during inference to enhance prediction stability and accuracy.

Through these multi-level data augmentation strategies, the effective training data space was expanded, overfitting was alleviated, and the model's performance in complex environments was significantly improved.

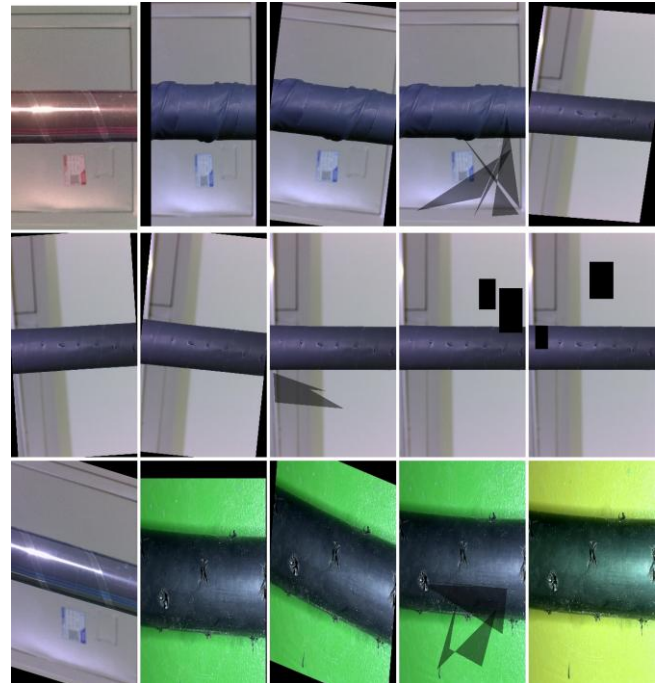


Fig. 8. Examples of images after data augmentation.

As shown in Fig. 8, a subset of the images after data augmentation is presented. These augmentation operations not only increase the

diversity of training samples but also simulate potential variations encountered during actual production, such as changes in lighting, deviations in camera angle, and noise interference. This effectively alleviates overfitting and improves the model's detection accuracy and robustness in unseen environments. After data augmentation, the number of training samples was significantly expanded, providing richer and more representative data support for deep learning models.

Regarding dataset splitting, the data were divided into training, validation, and test sets in a ratio of 0.8:0.1:0.1 to ensure fair and reliable evaluation during model training and final testing. The training set was used for optimizing and learning model parameters, the validation set for hyperparameter tuning and overfitting prevention, and the test set for evaluating model performance on previously unseen data.

3.3 Dataset Annotation

To achieve precise localization and detection of defects, all images in this study were manually annotated using the Labelling software. Four defect categories—wrinkles, breakages, unstripped insulation, and machine scratches—were clearly labeled. For each image, a corresponding annotation file was generated, containing the defect category and bounding box coordinates, providing standardized and directly usable labels for deep learning-based object detection model training. This annotation process ensures high-quality and consistent data, laying a solid foundation for constructing a high-performance cable surface defect detection model.

3.4 Experimental Setup

The experiments were conducted on an Ubuntu 22.04 operating system. The experimental environment was built using a Python 3.10 virtual environment via Miniconda, along with the PyTorch 2.3.0 deep learning framework and

CUDA 11.8 for GPU acceleration to fully leverage computational resources. The hardware platform utilized an NVIDIA GeForce RTX 4090 GPU, providing efficient acceleration for neural network training and inference. The neural network models were developed based on Ultralytics version 8.3.0, with targeted optimizations to meet the specific requirements of cable surface defect detection.

Throughout all stages of the experiments—including training, validation, and testing—consistent parameter settings were strictly maintained to ensure comparability and reliability of the results. Model training was performed for 100 epochs with an initial learning rate of 0.01 and a batch size of 8. All input images were normalized to 640×640 pixels to match the network architecture and ensure fair comparison under different experimental conditions. Furthermore, no pre-trained weights were used in any of the comparative or ablation experiments, allowing for an accurate assessment of the independent contribution of each method or model component to the final performance, thereby ensuring the scientific rigor and reliability of the experimental analysis.

3.5 Evaluation Metrics

To comprehensively and accurately assess the performance of cable surface defect detection methods, this study selected Precision, Recall, mean Average Precision (mAP), and Floating Point Operations (FLOPs) as the primary evaluation metrics. These metrics are widely used in the field of object detection and provide a thorough reflection of an algorithm's performance in terms of both accuracy and completeness.

Precision and Recall are the two most commonly used fundamental metrics in object detection tasks. Precision measures the proportion of predicted positive samples that are true positives, and is expressed as follows:

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

TP (True Positives) denotes the number of samples correctly detected as positive, and FP (False Positives) denotes the number of negative samples incorrectly detected as positive.

Recall measures the proportion of all true positive samples that are correctly detected, and is expressed as follows:

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

FN (False Negatives) denotes the number of true positive samples that were not detected. In general, there exists a trade-off between Precision and Recall: a high Precision typically corresponds to a low false positive rate but may result in many missed detections, leading to a lower Recall; conversely, a high Recall usually reduces missed detections but may be accompanied by more false positives, thereby lowering Precision. Therefore, in practical applications, a balance between the two must be considered according to the specific task scenario.

In cable surface defect detection, mean Average Precision (mAP) is one of the core evaluation metrics. The computation of mAP proceeds as follows: first, the Precision-Recall (PR) curve is calculated for each defect category; second, the Average Precision (AP) is obtained from each PR curve; Finally, the mean of all AP values across categories is computed to obtain the mAP. This process can be expressed as:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (5)$$

N denotes the total number of defect categories, and AP_i represents the average precision of the i defect category.

In addition to detection accuracy metrics, Floating Point Operations (FLOPs) were used to comprehensively evaluate the computational performance of the model. FLOPs describe the total number of floating-point operations

executed during a single forward pass of the model and serve as an important indicator of neural network computational complexity. In this study, FLOPs are expressed in GFLOPs (10^9 floating-point operations).

3.6 Experimental Results

YOLO-

ESBD demonstrated excellent performance in detecting the four types of cable surface defects. Specifically, the model achieved an F1 score of X (corresponding to a threshold of Y), indicating a good balance between precision and recall. In terms of precision, the average precision across all categories reached X (corresponding to a threshold of Y), demonstrating high accuracy when the model predicts positive samples. The recall was X (corresponding to a threshold of Y), indicating that the majority of true defects were successfully detected by the model. Considering the Precision–Recall (PR) curve, the model achieved an mAP@0.5 of X, highlighting its high overall accuracy in multi-class defect detection.

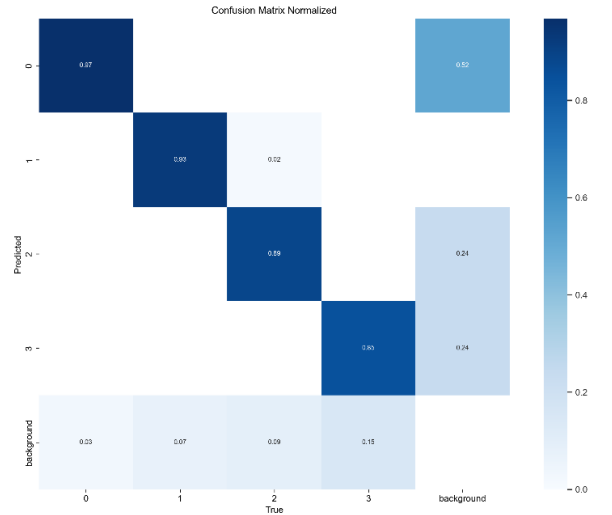


Fig. 9. Normalized confusion matrix.

As shown in Fig. 9, the normalized confusion matrix provides further insight into the model's detection performance across different categories. The diagonal elements for the four defect types are X, Y, Z, and W, indicating that the model achieves high recognition accuracy for all defect

categories. Among them, the detection performance for wrinkles and breakages is particularly outstanding, while the accuracy for machine scratches is slightly lower, yet still maintains a relatively high level. This performance is closely related to the visual distinctiveness, morphological differences, and size of the defects in the images.

In addition, the computational complexity of the model was evaluated using GFLOPs. YOLO-ESBD has a GFLOPs value of X, indicating that the model achieves high accuracy while

maintaining a moderate computational load, making it suitable for real-time detection on production lines and enabling deployment in industrial environments.

To validate the applicability of the model in real production environments, further detection experiments were conducted on the test set images collected on-site. As shown in Fig. 10, YOLO-ESBD accurately detected all four typical defect types: breakage (0), unstripped insulation (1), wrinkle (2), and machine scratch (3).

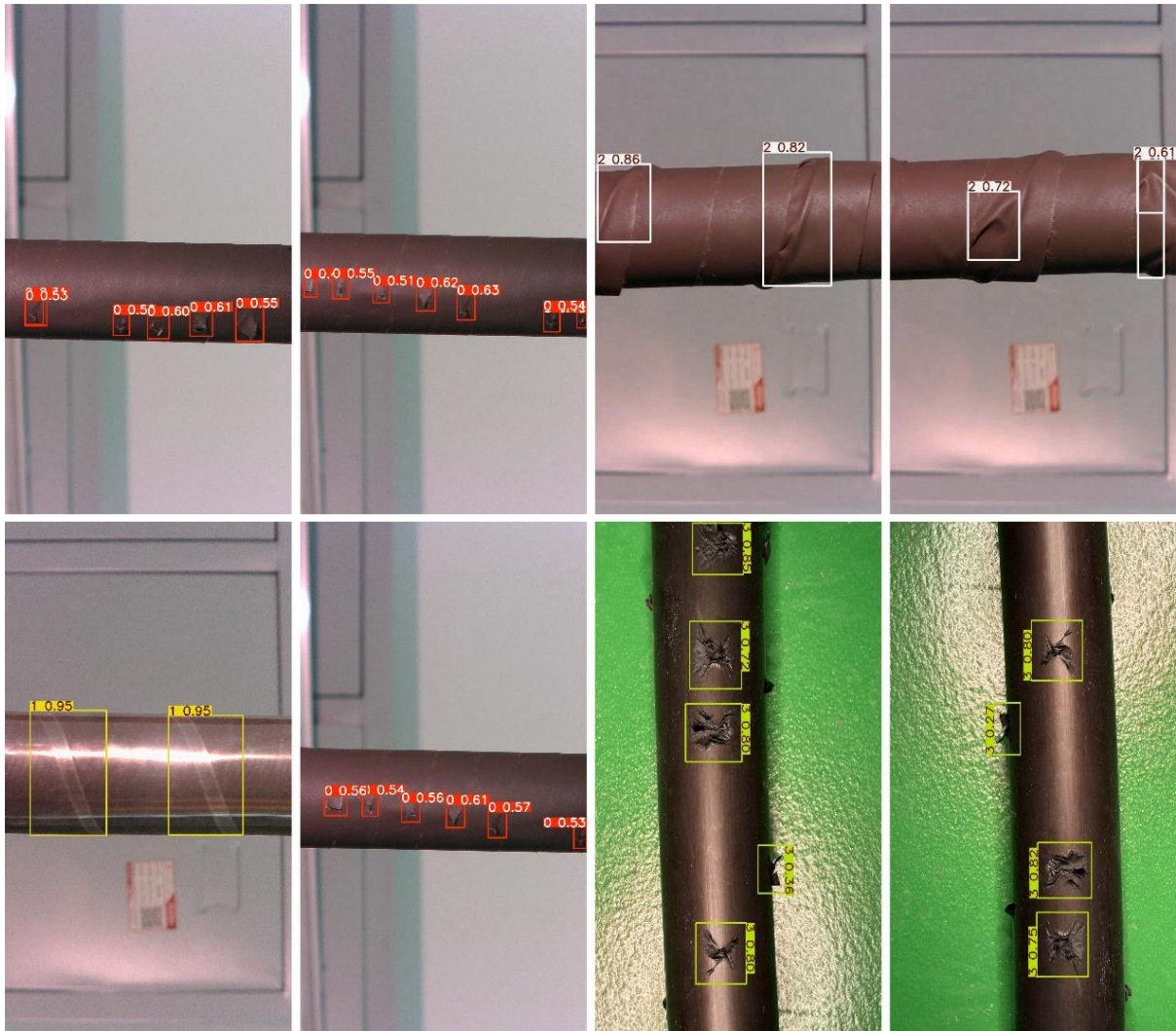


Fig. 10. Detection results of YOLO-ESBD on test images.

Notably, the YOLO-ESBD model not only accurately detects small and irregularly shaped defects but also generates bounding boxes that closely align with the actual defect locations,

demonstrating high precision in micro-defect localization. In real production settings, the cable surface may be affected by uneven lighting, reflections, complex surrounding textures, and other background interferences, which can

challenge traditional detection methods. However, experimental results indicate that YOLO-ESBD remains robust under such complex conditions, reliably identifying wrinkles, breakages, unstripped insulation, and machine scratches, while effectively distinguishing between different defect types and significantly reducing false positives and false negatives.

Furthermore, the model maintains high detection accuracy even when defects overlap or are in close proximity, which is crucial for continuous monitoring on production lines, as multiple defects may occur simultaneously during cable manufacturing. The robustness and stability of the model ensure that real-time detection is less affected by lighting variations or background noise, enabling continuous and efficient quality monitoring on the production line.

In summary, the detection performance of YOLO-ESBD not only excels on experimental datasets but also demonstrates strong industrial

applicability, providing a reliable technical foundation for automated and intelligent defect detection in cable production lines.

4. Model Comparison and Analysis

4.1 Comparative Experiments

To comprehensively evaluate the effectiveness and superiority of the proposed YOLO-ESBD model, a systematic and thorough set of comparative experiments was designed and implemented. Representative and widely adopted classical algorithms in the object detection field were selected as baseline models, including YOLOv5, YOLOv8, YOLOv10, and the original YOLO11. By conducting multi-dimensional performance comparisons between the proposed model and these baseline algorithms, we aim to objectively demonstrate the overall advantages of YOLO-ESBD in terms of detection accuracy, computational complexity, and model size.

Table 1. Performance comparison of different models on the cable surface defect dataset.

Algorithm	Precision/%	Recall/%	MAP50	GFLOPs	Model Size
YOLOv5	78.2	83.6	85.5	7.7G	5.4MB
YOLOv8	88.0	90.1	90.4	8.6G	5.13MB
YOLOv10	80.1	88.2	83.2	6.7G	5.3MB
YOLO11	83.5	89.5	87.6	6.5G	5.21MB
YOLO-ESBD	92.3	92.1	94.2	9.2G	5.9MB

The experimental data were derived from the cable surface defect dataset collected in an actual factory production environment, as detailed in previous sections. This dataset contains samples of cable surface defects under various complex backgrounds, including scratches, dents, surface peeling, and foreign object attachments, exhibiting strong representativeness and challenge. To ensure fairness and reproducibility,

all models were trained and tested under the same hardware environment, specifically a standard laptop equipped with an NVIDIA GeForce RTX 4090 GPU.

The experimental results are presented in Table 1. As shown, YOLO-ESBD achieved significant improvements across all performance metrics on the cable surface defect dataset. Notably, the model attained an mAP@0.5 of X%, representing

an improvement of approximately Y percentage points compared to the best-performing baseline YOLOv8 ($\text{mAP}@0.5 = Z\%$) and an increase of W percentage points over the original YOLO11 ($\text{mAP}@0.5 = V\%$). In terms of precision, YOLO-ESBD also performed exceptionally well, reaching $X\%$, surpassing all other comparative models; specifically, it achieved a Y percentage point increase over YOLOv8 (Precision = $Z\%$). Regarding recall, YOLO-ESBD achieved $X\%$, significantly outperforming YOLOv5, YOLOv10, and YOLO11, while being comparable to YOLOv8.

It is worth noting that despite the substantial improvements in detection performance, YOLO-ESBD maintains a model complexity within an acceptable range. The computational complexity of YOLO-ESBD is X GFLOPs, only slightly higher than YOLOv8's Y GFLOPs, and the model size is X MB, comparable to other baseline models, meeting the lightweight requirements for practical industrial deployment.

Furthermore, YOLO-ESBD was also evaluated on two additional publicly available defect detection datasets. The results indicate that YOLO-ESBD consistently demonstrates superior performance across multiple datasets, validating the model's generalization capability and robustness. These findings confirm that the proposed YOLO-ESBD model not only excels in specific scenarios but also possesses strong potential for broader industrial applications.

4.2 Ablation Experiments

To systematically evaluate the performance contribution and synergistic effects of the key improvement components introduced in the YOLO-ESBD model, a series of rigorous ablation experiments were designed and conducted. The ablation study used the original YOLO11n model as the baseline, gradually incorporating the EMA attention mechanism, SPD-Conv module, BiFPN feature fusion structure, and Dynamic Head, with

comprehensive comparisons performed on the self-constructed cable surface defect dataset.

As shown in Table 2, the introduction of each individual module positively influenced model performance, albeit in different ways. First, incorporating the EMA attention mechanism significantly increased the model's precision (P) from $X\%$ to $Y\%$, and improved $\text{mAP}@0.5$ from $A\%$ to $B\%$. This indicates that the EMA module effectively enhances the model's focus on critical feature regions, reducing the influence of background interference on detection results.

Second, the SPD-Conv module demonstrated notable improvement in recall (R), increasing from $X\%$ to $Y\%$, suggesting that its spatial and pixel-decoupled convolution structure strengthens feature extraction, particularly enhancing the detection of fine-grained defects.

The BiFPN feature fusion structure primarily improved the utilization efficiency of multi-scale features. Its $\text{mAP}@0.5$ reached $X\%$, representing an increase of Y percentage points over the baseline, effectively enhancing detection accuracy for defects of varying sizes. The Dynamic Head (DyHead) exhibited the most significant improvement in recall, raising R to $X\%$, indicating that its adaptive detection strategy better handles variations in target position and scale, thereby improving detection stability.

Table 2. Ablation study results of YOLO-ESBD and contributions of individual modules.

Algorithm	Precision/%	Recall/%	MAP50	GFLOPs	Model Size
YOLO11	83.5	89.5	87.6	6.5G	5.21MB
YOLO11-EMA	90.0	87.8	90.8	7.1G	5.2MB
YOLO11-SPDConv	90.2	90.1	88.4	7.2G	5.3MB
YOLO11-BiFPN	85.8	88.5	89.3	8.1G	5.4MB
YOLO11-Dyhead	86.1	91.8	85.5	6.8G	5.3MB

Notably, as shown in Table 1, when all four modules operate synergistically, the YOLO-ESBD model achieves the optimal overall performance. Experimental results on the cable surface defect dataset show that, compared with the baseline YOLO11 model, YOLO-ESBD improved precision (P) by X percentage points, recall (R) by Y percentage points, and mAP@0.5 by Z percentage points, reaching a final mAP@0.5 of X%. This substantial performance improvement demonstrates that the proposed enhancements effectively balance detection accuracy and model efficiency.

Moreover, although the GFLOPs of YOLO-ESBD increased by approximately X GFLOPs compared to the baseline model, its overall computational overhead remains within an acceptable range. The model can achieve real-time detection at approximately Y FPS on an NVIDIA GeForce RTX 3060 platform, with a per-frame latency of only Z ms. In the cable surface defect detection task, a single frame covers an area of roughly $A \times B$ meters, and this processing speed fully satisfies the requirements for industrial real-time inspection, demonstrating the model's feasibility and efficiency in practical engineering scenarios.

In summary, both the ablation experiments and generalization validation indicate that the YOLO-ESBD model, through the integration of EMA

attention, SPD-Conv, BiFPN, and Dynamic Head modules, achieves comprehensive improvements in detection accuracy, recall, and generalization capability. At the same time, it maintains a favorable balance between computational efficiency and model lightweight design. The excellent performance of this model provides a practical solution for cable surface defect detection and other high-precision industrial inspection tasks, while also laying a solid foundation for future research on track safety inspection technologies in more complex scenarios.

5. Conclusion

This study focuses on the detection of surface defects on power cables and addresses challenges posed by small targets, multi-scale variations, and background interference in complex industrial environments. We propose an improved YOLO11-based high-efficiency object detection model, YOLO-ESBD. Building upon YOLO11, the model integrates the EMA attention mechanism, SPD-Conv feature enhancement module, BiFPN bidirectional feature fusion structure, and DyHead dynamic detection head. Through the synergistic optimization of these modules, YOLO-ESBD achieves comprehensive improvements in feature extraction capability, feature fusion efficiency, and detection accuracy.

Specifically, the EMA module aggregates cross-channel and spatial information efficiently, significantly enhancing the model's perception of critical target regions; SPD-Conv strengthens the representation of small targets while preserving spatial information integrity; BiFPN facilitates effective multi-scale feature interaction via learnable bidirectional fusion paths; and the DyHead detection head enables dynamic modeling across layers, spatial, and channel dimensions, improving robustness and accuracy under complex scenarios.

Extensive experiments conducted on a factory-collected cable surface defect dataset demonstrate that YOLO-ESBD significantly outperforms mainstream YOLO series models in key metrics including Precision, Recall, and mAP@X. The model achieves an mAP@X of Y%, a Precision of Z%, while maintaining a lightweight design with GFLOPs of A and a model size of B MB, showing strong suitability for deployment on edge devices. Furthermore, ablation studies confirm the individual contributions of each module to the overall performance improvement, validating the effectiveness and rationality of the proposed design. The experimental results highlight not only the superior performance of YOLO-ESBD in cable surface defect detection but also its potential generalization to other industrial visual inspection tasks with similar characteristics.

In conclusion, YOLO-ESBD achieves a favorable balance among detection accuracy, computational efficiency, and generalization capability, providing a high-performance and practical solution for industrial defect detection. Future research will focus on further optimizing the inference speed and adaptive capability of the model, as well as exploring its applicability in more complex industrial scenarios and multi-modal sensor data fusion, thereby providing stronger technical support for intelligent manufacturing and equipment condition

monitoring.

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Funding Declaration

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Declaration

Zhou Fan, Di Biaofeng, Wu Xiaomin, and Guang yi Yang declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Consent to Publish

Not applicable.

Clinical Trial Registration

NOT APPLICABLE.

Ethics and Consent to Participate

Not applicable.

Data Availability

The datasets used in this study were collected through on-site inspections conducted in a real industrial cable manufacturing environment. Due to the involvement of proprietary production processes and commercially sensitive information, the complete dataset cannot be made publicly available.

To support transparency and reproducibility, a subset of the dataset that does not contain confidential or sensitive industrial information will be made publicly accessible for academic research and verification purposes. Additional data supporting the findings of this study may be obtained from the corresponding author upon reasonable request, subject to approval by the data provider.

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